Signal Rise Time and Transmission Line Length - Leo Marcinkus

Simulation Setup

The 50 Ω optimized microstrip transmission line (w_mils = 118 mil, verified by TDR) was simulated in Keysight ADS using a lossless substrate.The output rise time (RT_out) was measured at v_scope for line lengths of 6, 12, 24, and 36 inches. The degradation factor is RT_out / RT_in.

Summary Table

Length	RT_in (10-90%)	RT_out (10-90%)	RT_line (estimated)	Degradation Factor
6 in	10 ps	72 ps	71 ps	7.2x
12 in	10 ps	123 ps	123 ps	12.3x
24 in	10 ps	275 ps	275 ps	27.5x
36 in	10 ps	373 ps	373 ps	37.3x

Transient Waveforms

The following plots show v_tdr (input, red) and v_scope (output, blue) for each line length. Markers at 10% and 90% voltage levels are used to measure rise times.

6-inch Transmission Line:

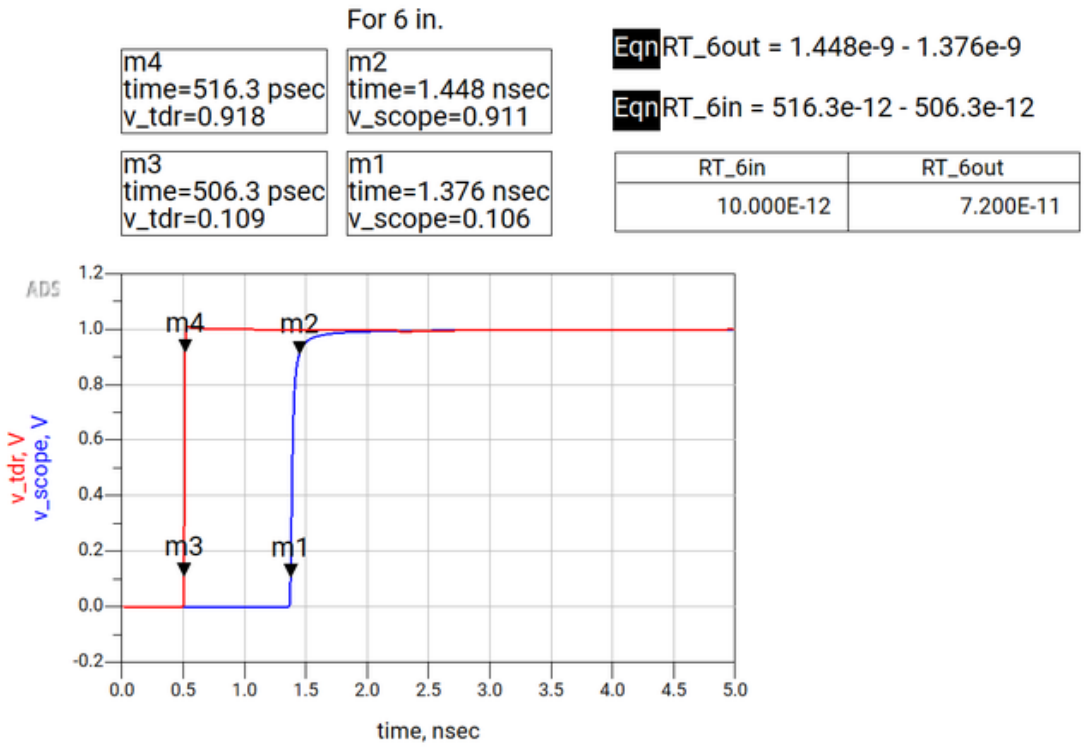


Figure 1: RT_in = 10 ps, RT_out = 72 ps (10-90%)

12-inch Transmission Line:

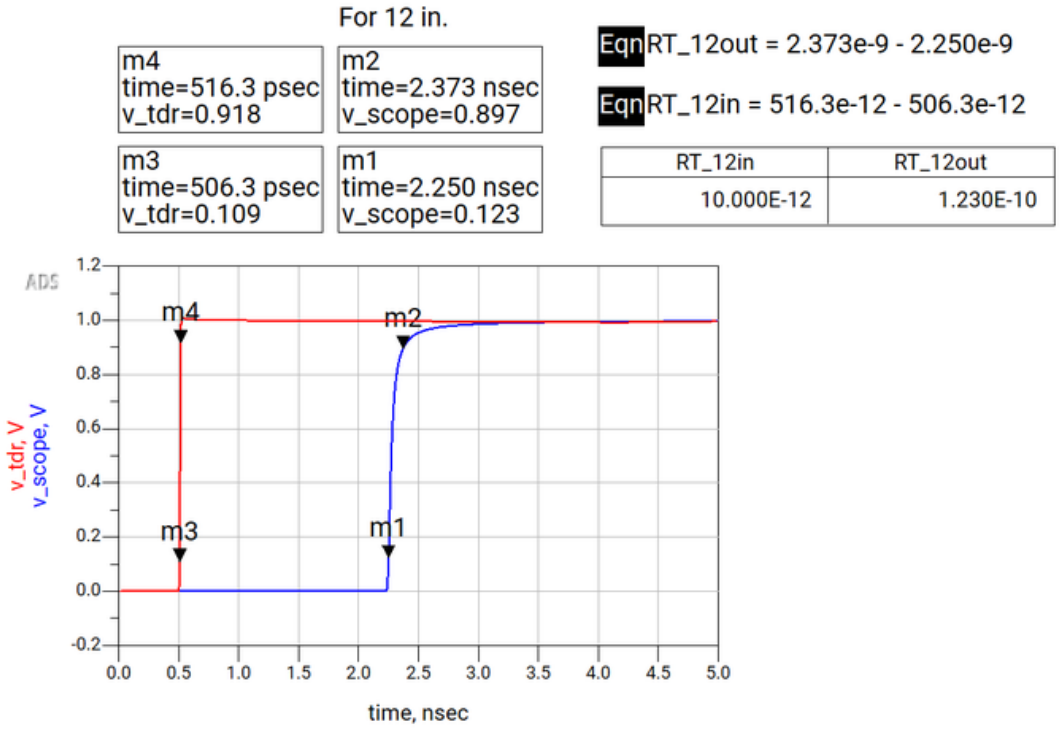


Figure 2: RT_in = 10 ps, RT_out = 123 ps (10-90%)

24-inch Transmission Line:

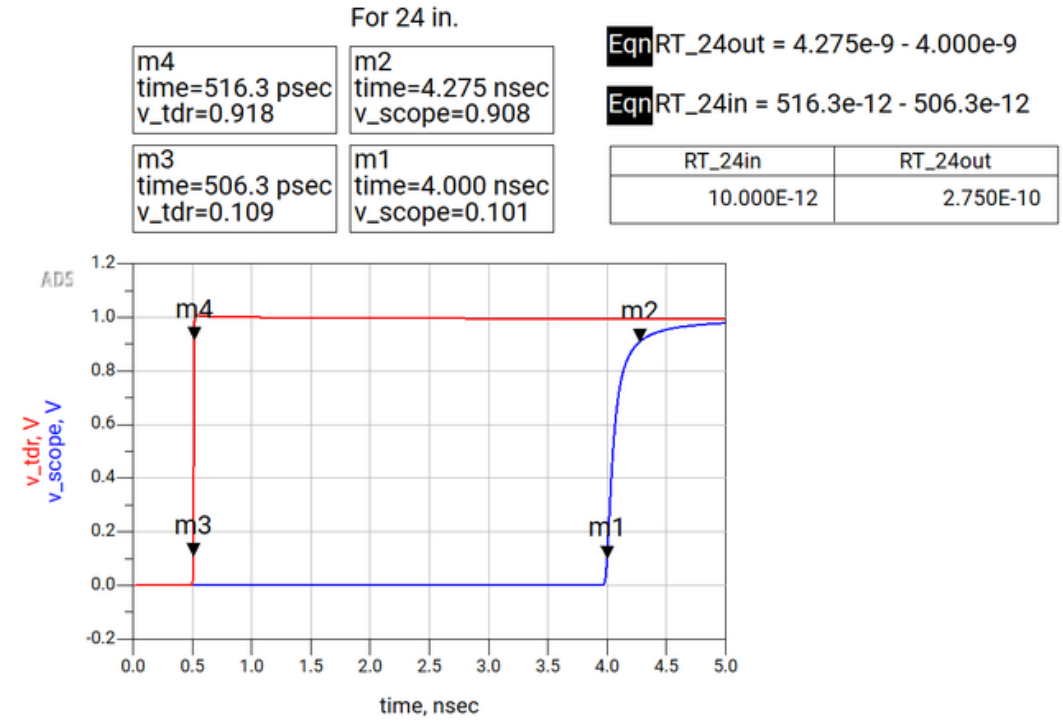


Figure 3: RT_in = 10 ps, RT_out = 275 ps (10-90%)

36-inch Transmission Line:

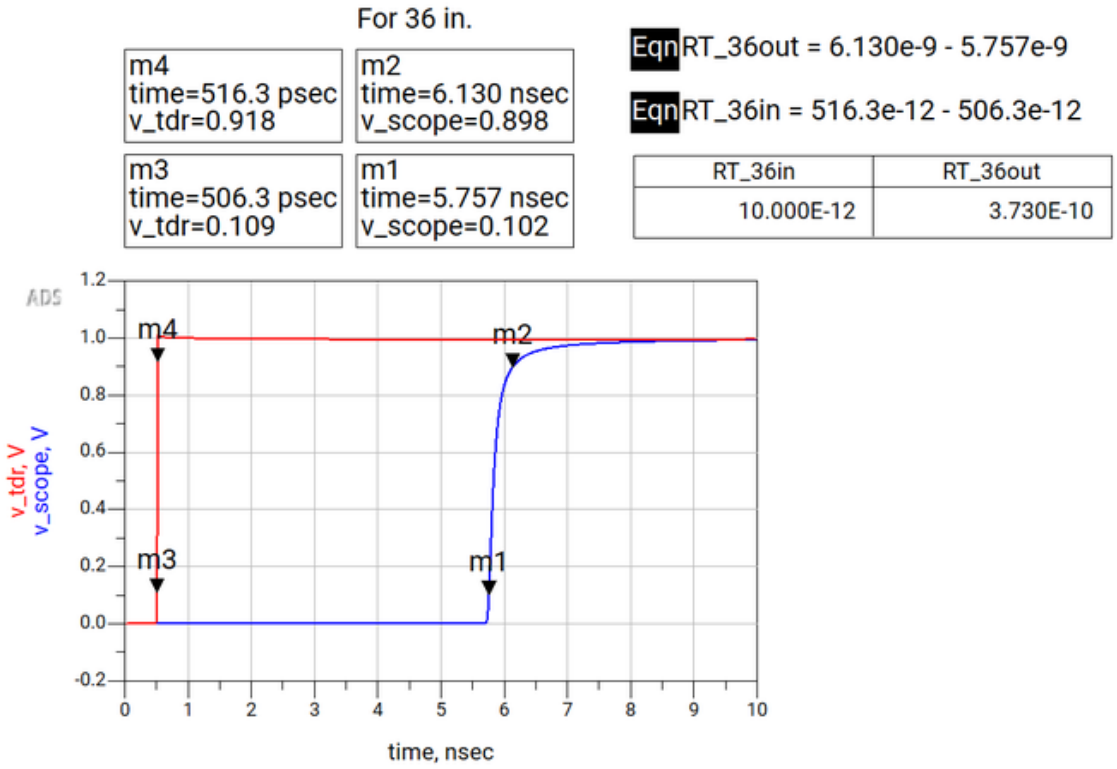


Figure 4: RT_in = 10 ps, RT_out = 373 ps (10-90%)